

SPAR Supply

The material availability specialist. Explains what stock exists, what has been ordered from suppliers and where material shortages threaten a plan or promise.

Made for: product, engineering, operations and business teams.
Reading time: about 12 minutes.
Truth standard: describes the repository as implemented on 1 August 2026.

SPAR

What SPAR is here to do

A role, not an all-powerful chatbot

In one sentence

Explains what stock exists, what has been ordered from suppliers and where material shortages threaten a plan or promise.

2

product areas grouped here

2

registered callable tools

4

allowed capability prefixes

0

agents it may delegate to

Receives

On-hand and warehouse position · Purchase-order and GRN status · AXLE material requirements · KILN component demand

Produces

Availability evidence · Shortage and supplier-risk explanation · Procurement/warehouse work item · Traceable source references

Allowed capability families

inventory.*

purchase.*

supplier.*

agent.*

The key distinction

The product-area grouping below shows where SPAR contributes across XELOR. The callable-tool table later shows the smaller, exact surface Agent OS can invoke today. Product ownership does not automatically create runtime authority.

The business areas under SPAR

Business ownership is broader than today's Agent OS tools

Purchase

Vendors, approval-bound purchase orders and goods receipts.

Inventory

Warehouses, stock balances and the append-only movement ledger behind the single stock write path.

How to read this grouping

SPAR is the named specialist when a mission needs facts or analysis from these areas. Each module still owns its own rules, records and write paths. The agent coordinates through registered services; it does not become the database owner.

What remains true underneath

- Each module keeps its own business rules and permissions.
- Cross-module work passes through narrow ports or registered capabilities.
- Tenant isolation, audit and workflow apply regardless of which agent is speaking.
- A readable agent answer never replaces the source record.

How SPAR participates in a mission

The mission engine controls order, parallel work and recovery



1 · Scope

ONYX turns the goal into a versioned graph run with a fixed tenant, user, limits and expected outputs.

2 · Authorise

The capability registry checks that SPAR is allowed to call the named tool; RBAC repeats the user permission check.

3 · Execute safely

The registered domain service performs the operation. The agent never receives a general SQL doorway.

4 · Preserve evidence

Inputs, outputs, node events and checkpoints are stored so the mission can be explained or recovered.

Delegation

SPAR is a specialist and does not delegate to other agents in the current registry.

What SPAR can call today

Every row is explicitly registered and permission-checked

Capability key	Mode	Result	User permission	Approval?
inventory.on-hand.read	Read	Stock on hand	inventory.stock.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

What “dispatch” means

If listed, `agent.action.dispatch` appends a governed work item for a named specialist. It does not directly change a sales order, stock balance, production order, journal, payment or customer message.

Read / analyse / simulate

Returns evidence or a reproducible view. No business record is changed.

Draft

Creates reviewable proposed content. A draft is never labelled as an approved or completed business outcome.

Execute

Effectful at the Agent OS level and therefore requires an approved ancestor node.

Permission

The agent's allow-list and the signed-in user's RBAC must both allow the call.

Who SPAR works with

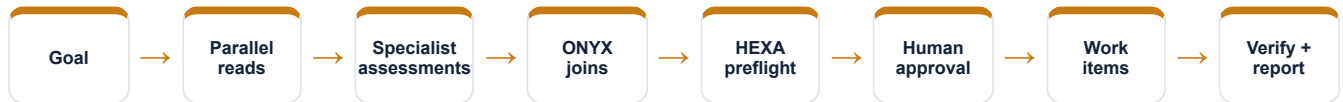
Specialists exchange evidence through ONYX, not hidden side conversations

AXLE Compares supply facts with planned material needs.

KILN Shows whether components can be issued for production.

RASP Exposes purchasing and inventory pressure affecting cash.

A complete controlled mission



Why this structure matters

Each agent stays accountable for its own facts. ONYX combines them, HEXA checks control evidence, and a person controls the decision boundary. This prevents one fluent answer from quietly becoming authority over the whole company.

Will material arrive before production needs it?

A realistic presentation path from question to controlled outcome

1

SPAR reads current on-hand stock. The wider Purchase module also holds approved orders and receipts, while Agent OS deliberately exposes a narrower live read surface today.

2

AXLE supplies the needed-by date and KILN supplies the production context; ONYX joins these facts without making SPAR guess outside its domain.

3

An approved expedite action becomes an append-only work item. A buyer still reviews and performs the actual supplier action.

What the presenter should say

“SPAR contributes verified domain evidence, with source references and limitations. It does not claim that a proposed or dispatched action is complete until the controlled system records and verifies the outcome.”

Useful questions to ask during the demo

- Which source records support SPAR's answer?
- What permission was checked?
- Is this a fact, a simulation, a draft or a completed result?
- Would this step wait for human approval?
- What happens if a node times out or the service restarts?

How SPAR stays trustworthy

Controls are enforced in the runtime and modules, not left to prompt wording

Control 1

Only Inventory owns the stock write path

Control 2

Balance rows are locked during posting and cannot go negative

Control 3

Stock movement history is append-only

Control 4

Purchase approval remains behind workflow

Identity boundary

Verified user + tenant

Authority boundary

Agent allow-list + user RBAC

Action boundary

Approval ancestry for side effects

Data boundary

Domain service + tenant-fenced database

Evidence boundary

Append-only run, event and audit records

What is live—and what is not

This page prevents the demo from overclaiming

Live in the repository

- On-hand evidence
- Warehouse and stock ledger foundation
- PO approval and GRN flow
- Supply contribution to cross-functional missions

Deliberate current limits

- Purchase-order read is present in the product but not yet an Agent OS capability
- No live supplier connector
- No autonomous PO creation or stock adjustment

Runtime disclosure

Orchestration, ERP reads, approval gates and governed action dispatch are live. Language reasoning is deterministic; no external model API or connector is active.

Primary implementation references

`apps/api/src/modules/inventory/`

`apps/api/src/modules/purchase/`

`apps/api/src/ports/stock.port.ts`

This guide is a plain-language operating map, not an API contract. The code, migrations and automated tests remain the binding technical source.

SPAR in one page

Use this page when explaining the agent to a new team member

Job

Explains what stock exists, what has been ordered from suppliers and where material shortages threaten a plan or promise.

Owns

Purchase, Inventory

Receives

On-hand and warehouse position · Purchase-order and GRN status · AXLE material requirements · KILN component demand

Returns

Availability evidence · Shortage and supplier-risk explanation · Procurement/warehouse work item · Traceable source references

Can call

inventory.on-hand.read, agent.action.dispatch

Cannot do

Purchase-order read is present in the product but not yet an Agent OS capability · No live supplier connector · No autonomous PO creation or stock adjustment

Plain-language glossary

Agent	A named specialist role with an allow-listed set of tools—not a free-roaming process.
Capability	One registered operation with an input, output, owner, mode and required permission.
Graph	A versioned mission recipe showing order, parallel steps, joins and approvals.
Checkpoint	A stored progress snapshot used for explanation and safe recovery.
Governed work item	An append-only assignment created after approval; it is not the final business mutation.